

Technical Document #77: Software Engineering - Comprehensive Overview

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Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Test-driven development (TDD) writes tests before writing code.
- API design defines how software components communicate.
- Refactoring improves code structure without changing behavior.